

Introduction

Background:

Environmental, Social, and Governance (ESG) metrics have become an increasingly important framework for evaluating corporate sustainability and long-term risk exposure. Investors and policymakers use ESG scores to assess whether firms are managing environmental impacts, social responsibility, and governance structures effectively. According to Yahoo Finance's ESG framework, lower ESG risk scores indicate lower exposure to sustainability-related risks.

At the same time, stock returns reflect financial performance and market valuation. Integrating ESG metrics with stock market data allows us to explore whether sustainability performance is associated with financial outcomes.

Statement of the Problem:

This project investigates whether ESG risk scores are associated with stock market performance and how ESG scores changed from 2024 to 2025. Specifically, we aim to:

- Examine the distribution and summary statistics of ESG scores.
- Compare ESG trends between 2024 and 2025.
- Analyze daily stock returns.
- Explore the relationship between ESG metrics and financial performance.

✓ Historical Stock Data

```
import pandas as pd
```

```
from google.colab import drive, files
drive.mount('/content/drive')
files.upload()
url_historical_stock = "/content/historical_stock_data_v3.csv"
historical_stock = pd.read_csv(url_historical_stock)
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `dr`

[Choose Files](#) historical_st...data_v3.csv

historical_stock_data_v3.csv(text/csv) - 11321407 bytes, last modified: 3/2/2026 - 100% done

Saving historical_stock_data_v3.csv to historical_stock_data_v3 (1).csv

✓ Historical Stock data : Ungrouped descriptive statistics

```
(
    historical_stock
    .info()
)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 188212 entries, 0 to 188211
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Symbol      188212 non-null object
1   Year         187844 non-null float64
2   Date         187844 non-null object
3   Open         187844 non-null float64
4   High         187844 non-null float64
5   Low          187844 non-null float64
6   Close        187844 non-null float64
7   Volume       187844 non-null float64
8   Dividend     2440 non-null  float64
9   Split        28 non-null   object
dtypes: float64(7), object(3)
memory usage: 14.4+ MB
```

```
stock_vars = ["Open", "High", "Low", "Close", "Volume", "Dividend"]
```

```
historical_stock[stock_vars].describe()
```

	Open	High	Low	Close	Volume
count	187844.000000	187844.000000	187844.000000	187844.000000	1.878440e+05
mean	170.386701	172.143744	168.550411	170.341924	4.789184e+06
std	420.399416	424.815038	415.982181	420.267384	1.816019e+07
min	1.010000	1.040000	0.970000	0.990000	5.100000e+03
25%	46.750000	47.247500	46.237500	46.727500	8.705000e+05
50%	91.570000	92.590000	90.590000	91.560000	1.774050e+06
75%	178.585000	180.480000	176.650000	178.550000	4.048275e+06
max	9914.170000	9964.770000	9794.000000	9924.400000	1.142269e+09

- Mean closing price is around 170.
- Large standard deviation (~420), indicating wide variation across firms.
- Volume is highly skewed, with maximum exceeding 1 billion shares.

The large dispersion in price and volume suggests that firms in the sample vary substantially in size and trading activity. The stock dataset is heterogeneous, containing both small-cap and large-cap firms.

✓ Historical Stock data : Grouped descriptive statistics

- By Date
- By Symbol

```
(
    historical_stock
    .groupby(["Date"])[["Open", "High", "Low", "Close", "Volume", "Dividend"]]
    .describe()
)
```

Date	Open								High
	count	mean	std	min	25%	50%	75%	max	cour
2024-01-02	604.0	154.892470	357.182667	1.79	43.2375	87.360	158.0000	6901.21	604
2024-01-03	604.0	154.438113	355.560891	1.87	43.9150	86.715	160.6025	6890.00	604
2024-01-04	604.0	153.166854	355.247192	1.80	43.4700	85.720	159.9275	6925.00	604
2024-01-05	604.0	152.866291	354.623554	1.72	43.3000	85.805	157.2925	6900.95	604
2024-01-08	604.0	153.212483	356.431344	1.70	43.6325	85.340	158.6000	6975.00	604
...	...		...	...		...	...	...	...
2025-03-24	604.0	170.943974	408.955644	1.57	47.2325	89.815	182.7475	7229.46	604
2025-03-25	604.0	172.594785	412.900907	1.56	47.4600	90.440	184.5650	7302.07	604
2025-03-26	604.0	172.526606	414.271412	1.47	47.7425	89.785	184.2375	7305.02	604
2025-03-27	604.0	171.761912	414.008970	1.42	46.9850	90.355	182.5350	7325.95	604
2025-03-28	604.0	171.689603	415.890986	1.46	46.9025	90.045	179.6775	7330.00	604

311 rows × 48 columns

```
(
    historical_stock
    .groupby(["Symbol"])[["Open", "High", "Low", "Close", "Volume", "Dividend"]]
    .describe()
)
```

	Open								High
	count	mean	std	min	25%	50%	75%	max	count
Symbol									
A	311.0	137.283183	7.593417	118.81	132.235	137.24	142.275	154.51	311
AA	311.0	35.863859	5.060669	25.91	31.925	36.00	39.785	47.24	311
AAL	311.0	13.533698	2.297036	9.10	11.320	13.92	14.930	19.09	311
AAP	311.0	55.527556	14.506504	33.50	42.135	58.85	65.740	86.71	311
AAPL	311.0	211.674920	25.214426	165.35	187.095	220.97	229.985	258.19	311
...	...	...	...	...	...	...	...	...	...
XYL	311.0	128.376817	8.197580	109.73	122.630	128.47	134.480	145.89	311
YUM	311.0	136.476367	7.399460	122.80	132.235	135.44	138.535	161.31	311
Z	311.0	59.893473	13.352651	39.05	48.920	56.39	72.730	87.89	311
ZBH	311.0	113.196431	8.188119	98.41	106.785	110.49	121.040	132.93	311
ZTS	311.0	176.794148	11.769369	146.25	168.255	175.70	186.630	200.50	311

621 rows × 48 columns

- Grouping by Date shows cross-sectional averages across firms.
- Grouping by Symbol reveals firm-level variation over time.

Daily averages are relatively stable across trading days, while firm-level statistics highlight strong cross-company differences in price levels and volatility.

✓ Stock Return

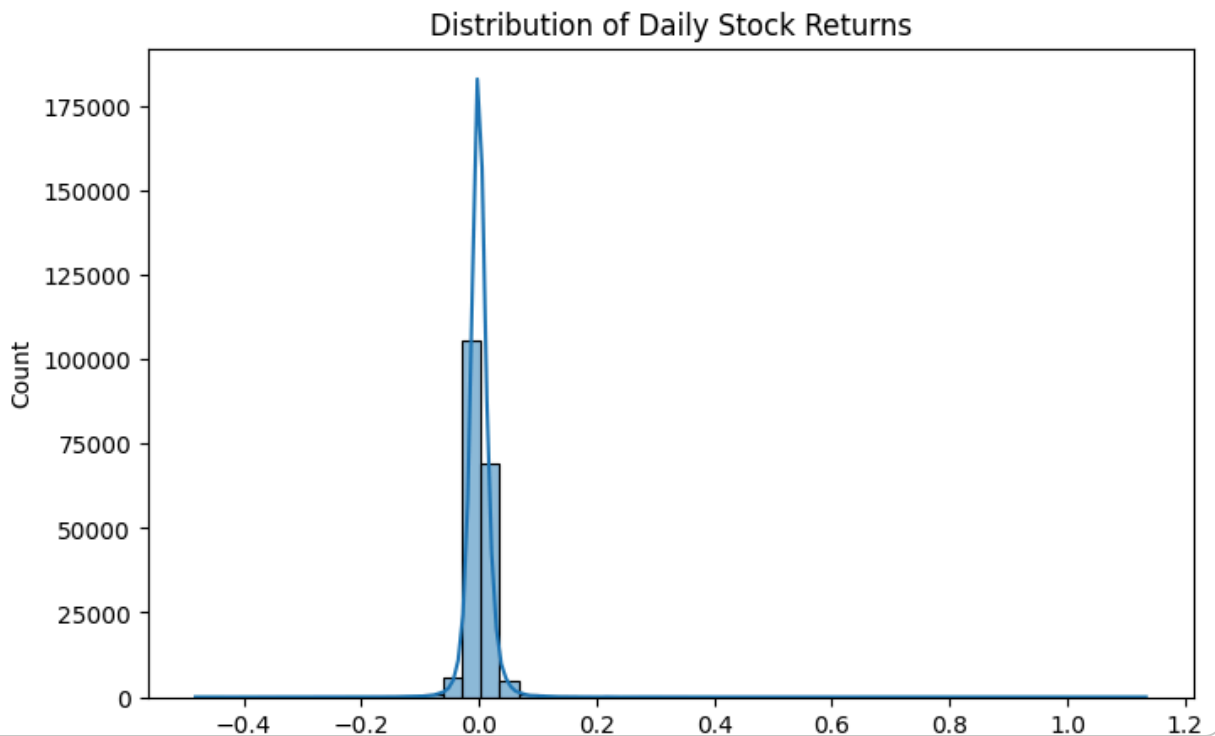
```
historical_stock["daily_return"] = historical_stock.groupby("Symbol")["Close"].
```

```
/tmp/ipykernel_1247/2863967872.py:1: FutureWarning: The default fill_method='ffill' is deprecated and will be removed in a future version.
historical_stock["daily_return"] = historical_stock.groupby("Symbol")["Close"]
```

✓ Stock Return : Distribution plot (Histogram)

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8,5))
sns.histplot(historical_stock["daily_return"].dropna(), bins=50, kde=True)
plt.title("Distribution of Daily Stock Returns")
plt.show()
```



- Returns are centered near zero.
- Distribution is sharply peaked.
- Presence of extreme outliers (fat tails).

From this financial return behavior, most daily returns are small, and extreme movements occur occasionally. The distribution is non-normal with heavy tails.

✓ ESG Data 2024

```
url_2024 = "https://bcdanl.github.io/data/esg_proj_2024_data.csv"
esg_proj_2024_data = pd.read_csv(url_2024)
esg_proj_2024_data
```

	Year	Symbol	Name	Sector	Industry	Country	Market_Cap
0	2024	A	Agilent Technologies Inc. Common Stock	Industrials	Biotechnology: Laboratory Analytical Instruments	United States	4036543481
1	2024	AA	Alcoa Corporation Common Stock	Industrials	Aluminum	United States	662213555
2	2024	AAL	American Airlines Group Inc. Common Stock	Consumer Discretionary	Air Freight/Delivery Services	United States	908802460
3	2024	AAP	Advance Auto Parts Inc.	Consumer Discretionary	Auto & Home Supply Stores	United States	447466529
4	2024	AAPL	Apple Inc. Common Stock	Technology	Computer Manufacturing	United States	261431000000
...	...	...	...	...	...	...	...
620	2024	XYL	Xylem Inc. Common Stock New	Industrials	Fluid Controls	United States	3201040268
621	2024	YUM	Yum! Brands Inc.	Consumer Discretionary	Restaurants	United States	3988504441
622	2024	Z	Zillow Group Inc. Class C Capital Stock	Consumer Discretionary	Business Services	United States	1019546912
623	2024	ZBH	Zimmer Biomet Holdings Inc. Common Stock	Health Care	Industrial Specialties	United States	2447677802
624	2024	ZTS	Zoetis Inc. Class A Common Stock	Health Care	Biotechnology: Pharmaceutical Preparations	United States	7253530835
625 rows × 13 columns							
Next steps: New interactive sheet							

✓ ESG 2024 : Ungrouped descriptive statistics

```
esg_vars = ["Total_ESG", "Environmental", "Social", "Governance", "Controversy"]
esg_proj_2024_data[esg_vars].describe()
```

	Total_ESG	Environmental	Social	Governance	Controversy	
count	625.000000	602.000000	602.000000	602.000000	573.000000	
mean	21.567040	5.747342	9.017774	6.819103	1.942408	
std	7.041637	5.200518	3.567490	2.371114	0.801113	
min	6.400000	0.000000	0.800000	2.400000	0.000000	
25%	16.300000	1.800000	6.700000	5.225000	1.000000	
50%	21.100000	3.950000	8.900000	6.300000	2.000000	
75%	26.000000	8.875000	11.175000	7.900000	2.000000	
max	52.000000	25.300000	22.500000	19.400000	5.000000	

ESG risk levels vary substantially across firms. Since lower scores indicate lower risk, firms at the lower end exhibit stronger sustainability profiles.

✓ ESG 2024 : Grouped descriptive statistics

By Sector and Industry

```
(
    esg_proj_2024_data
    .groupby(["Sector", "Industry"])[["Total_ESG", "Environmental", "Social", "C
    .describe()
)
```


		Total_ESG							
		count	mean	std	min	25%	50%	75%	
Sector	Industry								
Basic Materials	Forest Products	1.0	24.200000	NaN	24.2	24.200	24.20	24.2	
	Metal Mining	2.0	29.550000	2.899138	27.5	28.525	29.55	30.5	
	Precious Metals	5.0	18.140000	10.750953	7.3	8.200	20.50	21.4	
Consumer Discretionary	Advertising	1.0	16.300000	NaN	16.3	16.300	16.30	16.3	
	Air Freight/Delivery Services	5.0	27.140000	4.491436	19.7	26.400	28.60	30.3	
...	...	...	...	...	...	...	...	...	
Utilities	Natural Gas Distribution	7.0	29.228571	11.278108	19.0	22.650	23.60	32.3	
	Oil & Gas Production	1.0	25.100000	NaN	25.1	25.100	25.10	25.1	
	Oil/Gas Transmission	2.0	28.700000	8.343860	22.8	25.750	28.70	31.6	
	Power Generation	11.0	25.600000	2.794280	21.4	23.350	26.50	27.2	
	Water Supply	2.0	23.100000	2.121320	21.6	22.350	23.10	23.8	
138 rows × 40 columns									

Grouped descriptive statistics reveal:

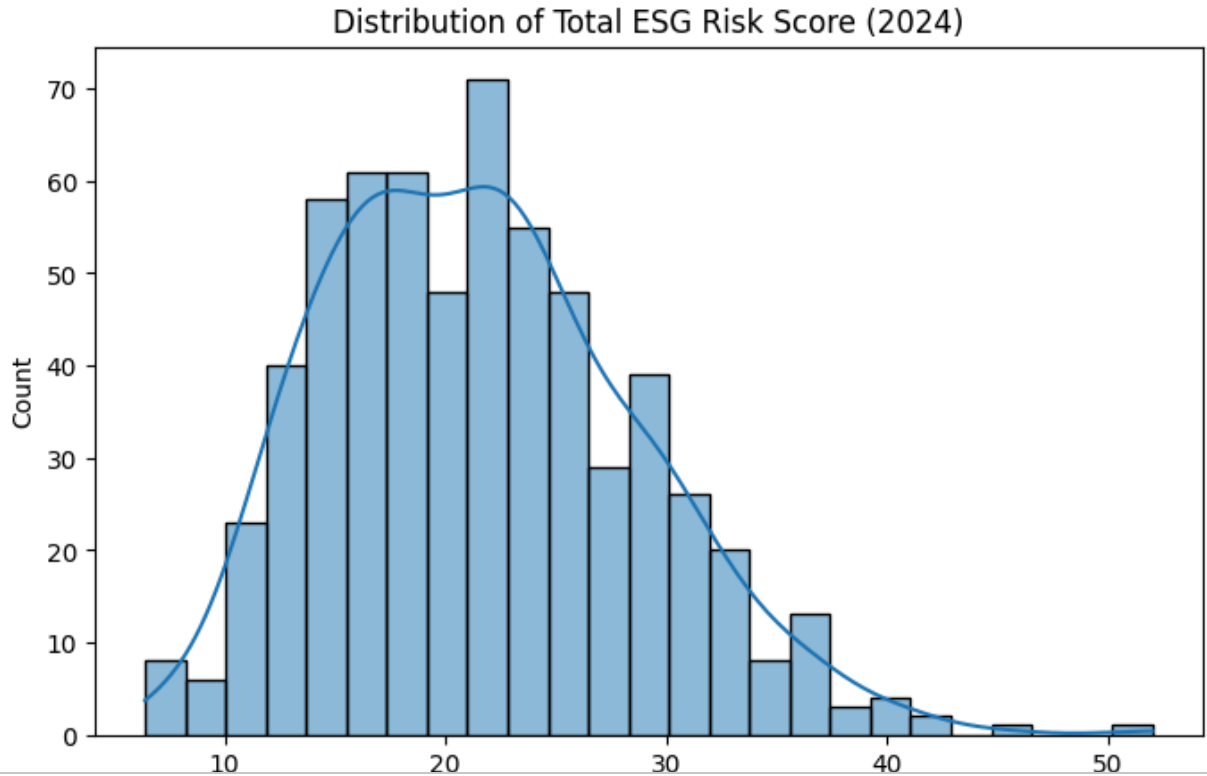
- Utilities and Basic Materials tend to have higher ESG risk exposure.
- Technology and some service industries show lower average ESG risk.

Sectoral differences are expected, as environmental exposure and regulatory pressures differ across industries.

✓ ESG 2024 : Distribution plot (Histogram)

```
plt.figure(figsize=(8,5))
sns.histplot(esg_proj_2024_data["Total_ESG"], kde=True, bins=25)
```

```
plt.title("Distribution of Total ESG Risk Score (2024)")
plt.show()
```



The 2024 Total ESG Risk Score distribution is approximately bell-shaped with moderate right skewness. Most firms cluster in the 15–25 range, indicating moderate ESG risk exposure. However, the distribution exhibits a noticeable right tail extending above 40, suggesting a small subset of firms face significantly higher sustainability-related risks. The dispersion of scores highlights meaningful heterogeneity in ESG performance across firms. Overall, ESG risk in 2024 is concentrated among a minority of firms, while the majority maintain moderate risk levels.

✓ ESG Data 2025

```
from google.colab import drive, files
drive.mount('/content/drive')
files.upload()
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount('/content/drive', force_remount=True)`

`esg_proj_2025.csv`

esg_proj_2025.csv(text/csv) - 62838 bytes, last modified: 3/1/2026 - 100% done

Saving `esg_proj_2025.csv` to `esg_proj_2025 (1).csv`

```
{'esg_proj_2025 (1).csv':
b'Symbol,total_esg,environmental,social,governance,controversy\r\nA,10.1,1.1,5.0
```

```
url_2025 = "/content/esg_proj_2025.csv"
esg_proj_2025 = pd.read_csv(url_2025)
esg_proj_2025
```

	Symbol	total_esg	environmental	social	governance	controversy	
0	A	10.1	1.1	5.0	3.9	1.0	
1	AA	25.1	14.7	8.0	2.4	3.0	
2	AADI	26.4	NaN	NaN	NaN	NaN	
3	AAL	23.7	9.6	11.3	2.8	2.0	
4	AAMI	31.3	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	...	
3142	FLYW	19.6	NaN	NaN	NaN	NaN	
3143	BIDU	18.0	2.9	8.1	7.0	3.0	
3144	SMBK	30.6	NaN	NaN	NaN	NaN	
3145	ESNT	22.3	NaN	NaN	NaN	NaN	
3146	KOP	24.1	NaN	NaN	NaN	1.0	

3147 rows × 6 columns

Next steps: [New interactive sheet](#)

```
esg_proj_2025["Year"] = 2025
esg_proj_2025
```

	Symbol	total_esg	environmental	social	governance	controversy	Year
0	A	10.1	1.1	5.0	3.9	1.0	2025
1	AA	25.1	14.7	8.0	2.4	3.0	2025
2	AADI	26.4	NaN	NaN	NaN	NaN	2025
3	AAL	23.7	9.6	11.3	2.8	2.0	2025
4	AAMI	31.3	NaN	NaN	NaN	NaN	2025
...	...	...	...	...	...	...	...
3142	FLYW	19.6	NaN	NaN	NaN	NaN	2025
3143	BIDU	18.0	2.9	8.1	7.0	3.0	2025
3144	SMBK	30.6	NaN	NaN	NaN	NaN	2025
3145	ESNT	22.3	NaN	NaN	NaN	NaN	2025
3146	KOP	24.1	NaN	NaN	NaN	1.0	2025

3147 rows × 7 columns

Next steps: [New interactive sheet](#)

```
(
    esg_proj_2025
    .info()
)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3147 entries, 0 to 3146
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Symbol          3147 non-null   object
1   total_esg       3147 non-null   float64
2   environmental    1158 non-null   float64
3   social          1158 non-null   float64
4   governance       1158 non-null   float64
5   controversy     1761 non-null   float64
6   Year            3147 non-null   int64
dtypes: float64(5), int64(1), object(1)
memory usage: 172.2+ KB
```

✓ ESG 2025 : Ungrouped descriptive statistics

```
esg_vars2 = ["total_esg", "environmental", "social", "governance", "controversy"]
```

```
esg_proj_2025[esg_vars2].describe()
```

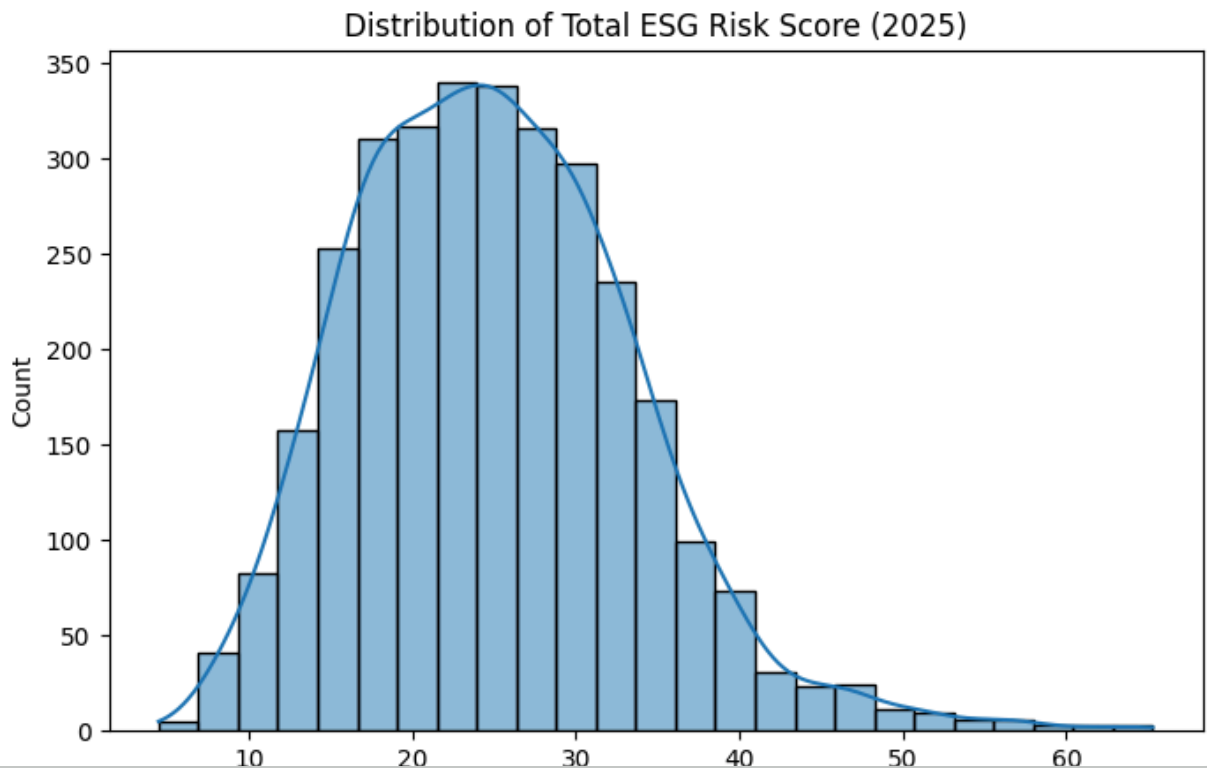
	total esg	environmental	social	governance	controversy
count	3147.000000	1158.000000	1158.000000	1158.000000	1761.000000
mean	25.005084	6.166149	9.505613	5.771071	1.582056
std	8.529579	5.511283	3.807264	2.529303	0.696743
min	4.500000	0.000000	0.300000	0.900000	1.000000
25%	18.600000	1.800000	7.100000	4.100000	1.000000
50%	24.500000	4.400000	9.400000	5.200000	1.000000
75%	30.400000	9.400000	11.800000	6.900000	2.000000
max	65.300000	32.500000	27.900000	18.600000	5.000000



The 2025 ESG scores show higher mean and greater dispersion than those of 2024. This suggests broader coverage of firms and/or increased ESG risk heterogeneity.

✓ ESG 2025 : Distribution plot (Histogram)

```
plt.figure(figsize=(8,5))
sns.histplot(esg_proj_2025["total_esg"], kde=True, bins=25)
plt.title("Distribution of Total ESG Risk Score (2025)")
plt.show()
```



The distribution of Total ESG Risk Scores in 2025 is approximately bell-shaped with slight right skewness. Most firms cluster in the 20–30 range, indicating moderate ESG risk exposure. However, the right tail extends beyond 50, suggesting a small subset of firms faces substantially higher sustainability-related risks. The dispersion of scores highlights significant heterogeneity across firms in ESG performance. Overall, while the majority of companies maintain moderate ESG risk levels, a minority exhibits elevated exposure that may warrant closer regulatory or investor attention.

✓ ESG 2024 + 2025

```
common_symbols = set(esg_proj_2024_data["Symbol"]).intersection(
    set(esg_proj_2025["Symbol"])
)
len(common_symbols)
```

625

```
esg_proj_2025 = esg_proj_2025.rename(columns={ "total_esg": "Total_ESG", "envir
```

```
esg_2024_common = esg_proj_2024_data[
    esg_proj_2024_data["Symbol"].isin(common_symbols)
]
```

```
esg_2025_common = esg_proj_2025[
    esg_proj_2025["Symbol"].isin(common_symbols)
]

esg_2024_common = esg_2024_common.drop_duplicates()
esg_2025_common = esg_2025_common.drop_duplicates()
```

```
esg_2024_2025 = pd.concat(
    [esg_2024_common, esg_2025_common],
    ignore_index=True
)

esg_2024_2025
```

	Year	Symbol	Name	Sector	Industry	Country	Market_Cap
0	2024	A	Agilent Technologies Inc. Common Stock	Industrials	Biotechnology: Laboratory Analytical Instruments	United States	4.036543e+10
1	2024	AA	Alcoa Corporation Common Stock	Industrials	Aluminum	United States	6.622136e+09
2	2024	AAL	American Airlines Group Inc. Common Stock	Consumer Discretionary	Air Freight/Delivery Services	United States	9.088025e+09
3	2024	AAP	Advance Auto Parts Inc.	Consumer Discretionary	Auto & Home Supply Stores	United States	4.474665e+09
4	2024	AAPL	Apple Inc. Common Stock	Technology	Computer Manufacturing	United States	2.614310e+11
...	...	...	...	...	...	...	...
1245	2025	ZTS	NaN	NaN	NaN	NaN	NaN
1246	2025	MD	NaN	NaN	NaN	NaN	NaN
1247	2025	PSA	NaN	NaN	NaN	NaN	NaN
1248	2025	SEIC	NaN	NaN	NaN	NaN	NaN
1249	2025	BIDU	NaN	NaN	NaN	NaN	NaN

1250 rows × 13 columns

Next steps:

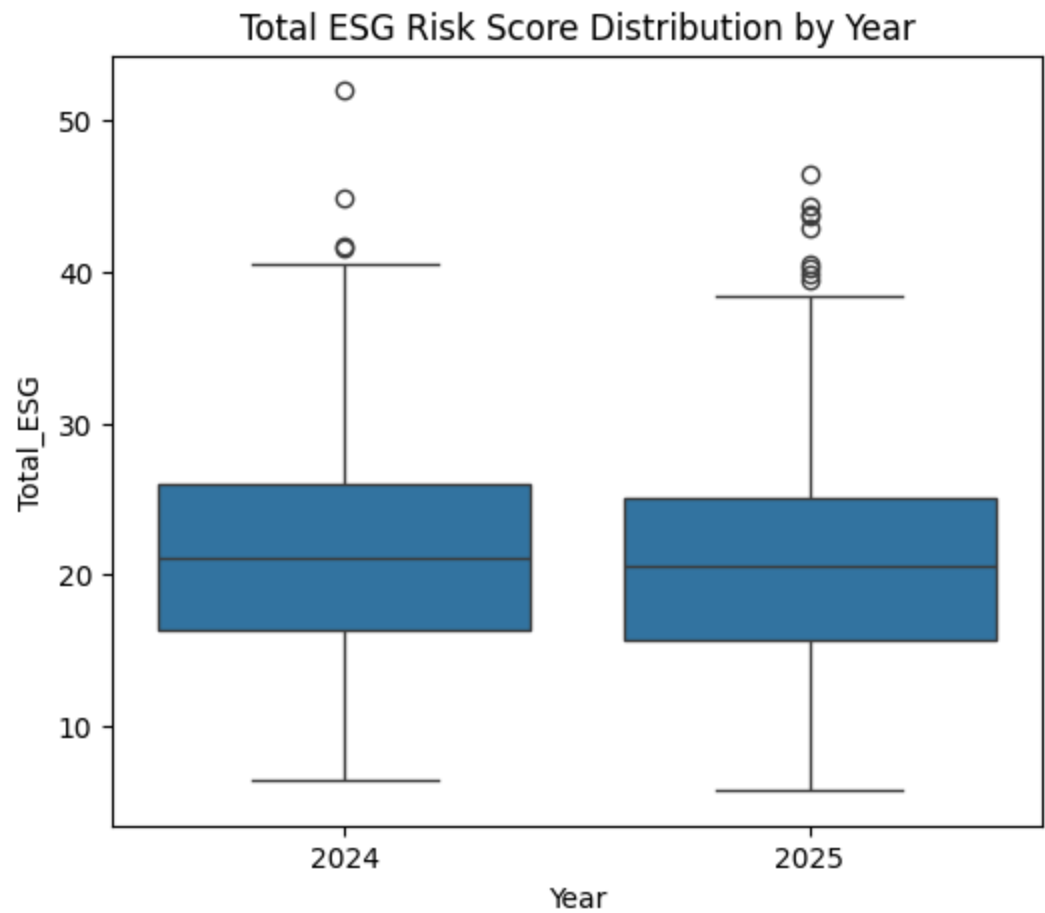
New interactive sheet

- ✓ Exploratory Data Analysis
- ✓ 1. Did the average ESG score change from 2024 to 2025?

```
esg_2024_2025.groupby("Year")["Total_ESG"].describe()
```

	count	mean	std	min	25%	50%	75%	max	
Year									
2024	625.0	21.56704	7.041637	6.4	16.3	21.1	26.0	52.0	
2025	625.0	20.83216	6.990241	5.7	15.6	20.5	25.0	46.5	

```
plt.figure(figsize=(6,5))
sns.boxplot(data=esg_2024_2025, x="Year", y="Total_ESG")
plt.title("Total ESG Risk Score Distribution by Year")
plt.show()
```



Mean ESG decreased from approximately 21.57 (2024) to 20.83 (2025).

On average, ESG risk exposure appears lower in 2025. This could reflect:

- Increased risk assessment stringency
- Changes in firm sustainability performance

✓

2. Did Environmental, Social, Governance or Controversy scores change differently?

```
component_means = esg_2024_2025.groupby("Year")[[
    "Environmental",
    "Social",
    "Governance",
    "Controversy"
]].mean().reset_index()
```

component_means

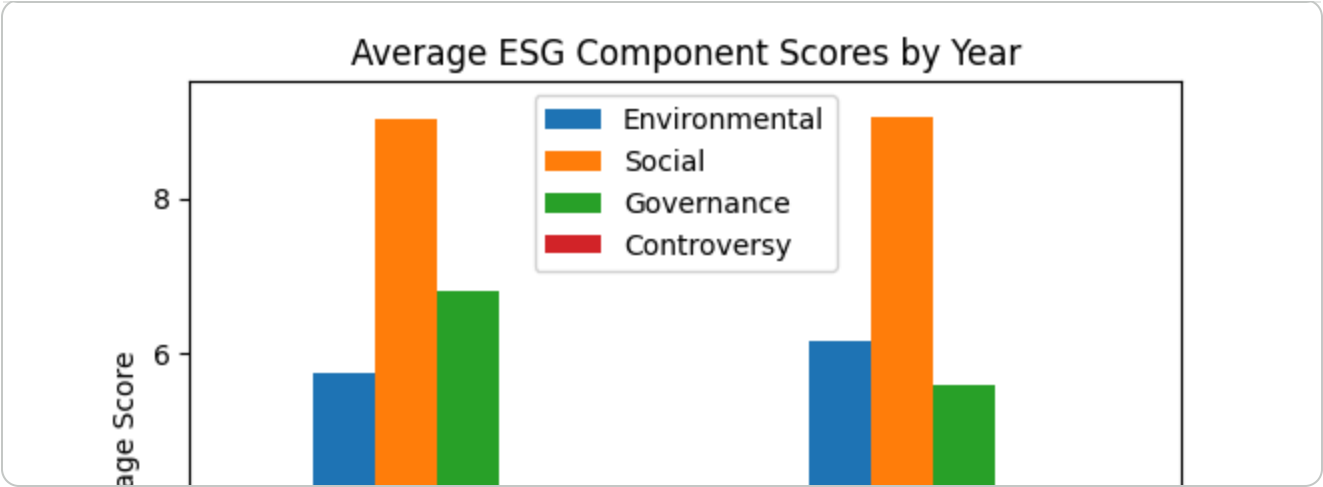
	Year	Environmental	Social	Governance	Controversy	
0	2024	5.747342	9.017774	6.819103	1.942408	
1	2025	6.176589	9.052508	5.606355	1.900178	

Next steps:

New interactive sheet

```
component_means.set_index("Year").plot(kind="bar")

plt.title("Average ESG Component Scores by Year")
plt.ylabel("Average Score")
plt.xlabel("Year")
plt.xticks(rotation=0)
plt.show()
```



The average ESG component scores reveal divergent trends between 2024 and 2025. Environmental risk increased from 5.75 to 6.18, indicating slightly greater exposure to environmental liabilities such as carbon emissions and climate-related risks. Social risk remained relatively stable and continues to represent the largest risk dimension among the components. In contrast, Governance risk declined from 6.82 to 5.61, suggesting improved corporate oversight, stronger board structures, and enhanced transparency practices. The Controversy score decreased slightly, implying marginal reductions in ESG-related incidents. Overall, governance structures appear to have strengthened, while environmental exposure increased modestly, highlighting the multidimensional nature of ESG risk.

3. Did ESG score distribution become more or less dispersed?

```
esg_2024_2025.groupby("Year")["Total_ESG"].std()
```

Total_ESG
